

Transformer Mechanics Worksheet — SET A

Question 1: Tokenization

(a) Split "UNHAPPINESS" into 3 reasonable sub-word pieces.

Answer:

UN + HAPPY + NESS

Justification:

- UN is a common prefix meaning *not* or *opposite of*.
- HAPPY is the root word that expresses the main meaning.
- NESS is a suffix that converts an adjective into a noun, representing a state or condition.

This split preserves the word's morphological structure, making it easier for the model to understand both meaning and grammar.

(b) Why does sub-word tokenization help?

Answer:

Sub-word tokenization enables a model to recognize meaningful parts of words instead of treating every complete word as unique. Even if the model has never seen "UNHAPPINESS" during training, it already understands the sub-words "UN," "HAPPY," and "NESS," allowing it to infer the meaning of the complete word accurately. This approach also reduces the vocabulary size and handles unseen words more effectively.

Question 2: Embeddings

(a) Which pair would have higher cosine similarity?

Answer:

The pair "cat" and "kitten" would have a higher cosine similarity.

Reason:

Both words are semantically related because a kitten is a young cat. They frequently appear in similar contexts, so their embedding vectors are located close together in the embedding space. In contrast, "cat" and "car" have unrelated meanings and therefore their vectors are much farther apart.

(b) What does an embedding represent, and how is it different from a one-hot vector?

Answer:

An embedding is a dense numerical vector that captures the semantic and contextual meaning of a word. Words with similar meanings have similar embeddings because they appear in similar contexts.

A one-hot vector, however, only identifies the word's position in the vocabulary. It contains mostly zeros with a single one and does not encode any information about meaning or relationships between words. Unlike embeddings, one-hot vectors cannot represent semantic similarity.

Question 3: Self-Attention Worked Example

Sentence:

"The trophy didn't fit in the suitcase because it was too big."

(a) The word "it" refers to:

Trophy

(b) Why is self-attention needed?

Answer:

Self-attention allows the model to examine relationships between all words in the sentence instead of relying only on their positions. In this sentence, understanding that "the trophy was too big" is the reason it could not fit into the suitcase requires semantic reasoning. A simple word-order rule cannot reliably determine the correct reference, whereas self-attention identifies the meaningful connection between "it" and "trophy."

Question 4: Query, Key, Value (5 Marks)

Answer:

In self-attention, every token is transformed into three vectors:

- **Query (Q):** Represents what the current word is searching for or asking about from other words in the sequence.
- **Key (K):** Represents the information each word offers for matching with other words' queries.
- **Value (V):** Contains the actual information or representation that will be passed to the next layer if the word receives attention.

The attention score is calculated by comparing the Query of one word with the Keys of all other words using a similarity measure (such as the dot product). These scores are normalized using the Softmax function to produce attention weights. The final output is obtained by computing the weighted sum of the Value vectors, allowing the model to focus on the most relevant words when generating contextual representations.